

Data Cleaning & Reporting Automation

1. Project Title

Data Cleaning & Reporting Automation using Python, Excel, and Power BI

2. Project Overview

Organizations often receive raw data containing missing values, duplicate records, inconsistent formats, and data entry errors. Cleaning such data manually is time-consuming and error-prone.

This project automates the complete data preprocessing workflow using Python and Excel, followed by automated reporting and dashboard creation in Power BI. The solution improves data quality, reduces manual effort, and provides actionable insights through automated reports and visualizations.

3. Problem Statement

Business data collected from multiple sources often contains:

- Missing values
- Duplicate records
- Inconsistent text formatting
- Invalid entries
- Data quality issues

These problems lead to inaccurate analysis and poor decision-making. A systematic automated solution is required to clean, validate, and report data efficiently.

4. Objectives

Primary Objectives

- Automate data cleaning processes.
- Handle missing values automatically.
- Remove duplicate records.
- Standardize inconsistent data formats.
- Generate automated reports.
- Create interactive dashboards.
- Improve overall data quality.

Secondary Objectives

- Reduce manual effort.

- Improve reporting efficiency.
 - Increase data reliability.
 - Support better business decisions.
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5. Tools and Technologies Used

Tool	Purpose
Python	Data Cleaning Automation
Pandas	Data Manipulation
NumPy	Data Processing
Excel	Data Storage
Power BI	Dashboard Development
ReportLab	PDF Report Generation

6. Dataset Description

Raw Dataset Fields

Column Name	Description
CustomerID	Unique customer identifier
Region	Customer location
Sales	Sales amount
Profit	Profit generated
OrderDate	Date of order

Raw Dataset Issues Introduced

Issue Type	Example
Missing Values	Sales column contains blanks
Duplicate Records	Same customer repeated
Inconsistent Text	North vs north
Null Categories	Missing region names

7. Data Cleaning Process

Step 1: Load Raw Dataset

The raw dataset is imported into Python using Pandas.

```
df = pd.read_excel("Raw_Dataset.xlsx")
```

Step 2: Handle Missing Values

Missing sales values are replaced with the median value.

```
df["Sales"] = df["Sales"].fillna(df["Sales"].median())
```

Reason:

- Median is less affected by outliers.
- Preserves data distribution.

Step 3: Handle Missing Categories

Missing regions are replaced with "Unknown".

```
df["Region"] = df["Region"].fillna("Unknown")
```

Step 4: Standardize Text Values

Convert inconsistent region names.

Example:

Before	After
north	North
NORTH	North
North	North

```
df["Region"] = df["Region"].replace({"north":"North"})
```

Step 5: Remove Duplicates

Duplicate records are removed.

```
df = df.drop_duplicates()
```

Step 6: Export Clean Dataset

```
df.to_excel("Cleaned_Dataset.xlsx")
```

8. Data Cleaning Results

Before Cleaning

Metric	Value
Total Records	210
Missing Values	7
Duplicate Rows	10
Inconsistent Categories	Present

After Cleaning

Metric	Value
Total Records	200
Missing Values	0
Duplicate Rows	0
Standardized Categories	Yes

9. Automated Reporting

The automation script generates a report after cleaning.

Report Includes

- Total records processed
- Missing values handled
- Duplicates removed
- Data quality improvements
- Processing summary

Sample Report Output

Metric	Value
Raw Records	210
Clean Records	200
Missing Values Fixed	7
Duplicates Removed	10
Status	Success

10. Power BI Dashboard Design

Page 1: Data Quality Overview

KPI Cards

- Total Records
- Clean Records
- Missing Values Fixed
- Duplicates Removed

Charts

- Missing Values by Column
 - Duplicate Records Removed
 - Data Quality Score
-

Page 2: Sales Analysis

KPI Cards

- Total Sales
- Average Sales
- Maximum Sales
- Minimum Sales

Visuals

- Sales by Region
 - Monthly Sales Trend
 - Top Customers by Sales
-

Page 3: Profit Analysis

KPI Cards

- Total Profit
- Average Profit
- Profit Margin

Visuals

- Profit by Region
 - Profit Trend
 - Loss-Making Transactions
-

Page 4: Regional Performance

Visuals

- Sales Distribution by Region
 - Profit Distribution by Region
 - Customer Count by Region
-

11. Automation Workflow

```
Raw Dataset
  ↓
Load Data
  ↓
Missing Value Treatment
  ↓
Data Standardization
  ↓
Duplicate Removal
  ↓
Generate Clean Dataset
  ↓
Generate Report
  ↓
Create Dashboard
```

12. Key Performance Indicators (KPIs)

KPI	Formula
Total Records	COUNTROWS()
Clean Records	COUNTROWS(Clean Data)
Missing Values Fixed	Calculated Count
Duplicates Removed	Raw – Clean
Total Sales	SUM(Sales)
Average Sales	AVERAGE(Sales)
Total Profit	SUM(Profit)
Profit Margin	Profit ÷ Sales

13. Benefits of the Project

Business Benefits

- Faster reporting
- Higher data accuracy
- Reduced manual effort
- Better decision-making

Technical Benefits

- Automated workflows
- Reusable Python scripts
- Scalable data processing
- Improved reporting efficiency

14. Project Deliverables

Deliverable	Status
Raw Dataset	✓
Clean Dataset	✓
Python Automation Script	✓
Automated Summary Report	✓
Power BI Dashboard Design	✓
Project Documentation	✓

15. Conclusion

The Data Cleaning & Reporting Automation project successfully automates the process of identifying and correcting data quality issues. Using Python, Excel, and Power BI, the solution reduces manual cleaning effort, improves data accuracy, and generates automated reports and dashboards. The project demonstrates practical

applications of data preprocessing, reporting automation, and business intelligence tools, making it highly relevant for data analyst and business analyst portfolios.